SON 2379: A National Framework for Predicting Nutrient Dynamics in River-Floodplain Restoration Projects

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Problem:

USACE invests in river and floodplain restoration, yet depending on flow conditions, residence time, and geomorphology, restored floodplains can function as sources or sinks that can trap or release nutrients, potentially causing habitat degradation. Planners lack a standardized toolkit for predicting this trade-off, risking designs that inadvertently damage ecosystems rather than improving them. ERDC's ClearWater-Riverine model, coupled with HEC-RAS 2D, provides foundational capabilities to evaluate nutrient cycling and impacts, but it has not yet been applied to produce nationally transferable guidance.

Envisioned Product:

Three integrated products: 1. a set of calibrated ClearWater-Riverine models for representative channel and floodplain geomorphic types found nationwide (braided channels, oxbow lakes, backwaters), providing a ready-to-use foundation for project-specific evaluations; 2. nutrient performance curves that translate complex model outputs into decision-relevant visualizations of the trade-off between hydrologic connectivity and nutrient retention or export; and 3. a national best practices guide with step-by-step protocols for applying the framework to assess nutrient dynamics in restoration design.

Value Added:

The framework will enable Districts to explicitly quantify whether a proposed restoration design will retain or export nutrients before implementation, transitioning nutrient impact assessment from ad hoc, site-specific judgment to a predictive, process-based approach grounded in validated models. By providing planners with generalized nutrient performance curves and a best practices guide, the framework ensures that restoration designs deliver net nutrient retention rather than unintended nutrient export. This will reduce risk and maximize ecosystem benefits.

Regional/National Significance:

The framework applies to every USACE District engaged in ecosystem restoration and management. The representative geomorphic types and hydrologic conditions will be selected to span the range of floodplain settings encountered across the national portfolio, facilitating transferability without extensive site-specific recalibration. The nutrient performance curves will provide the scientific basis for quantifying environmental benefits in Planning studies and benefit-cost analyses. The products will directly support the USACE ecosystem restoration mission and national goals for mitigating Harmful Algal Blooms, positioning restoration projects as verified nature-based solutions for nutrient management.